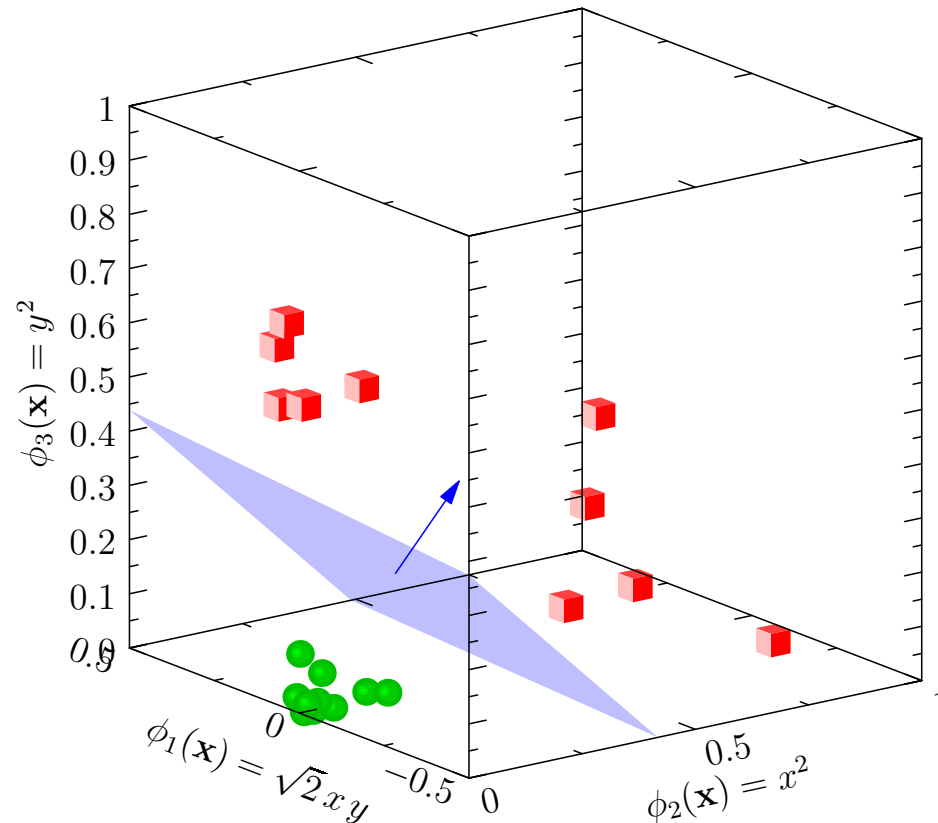


Mathematics of Machine Learning

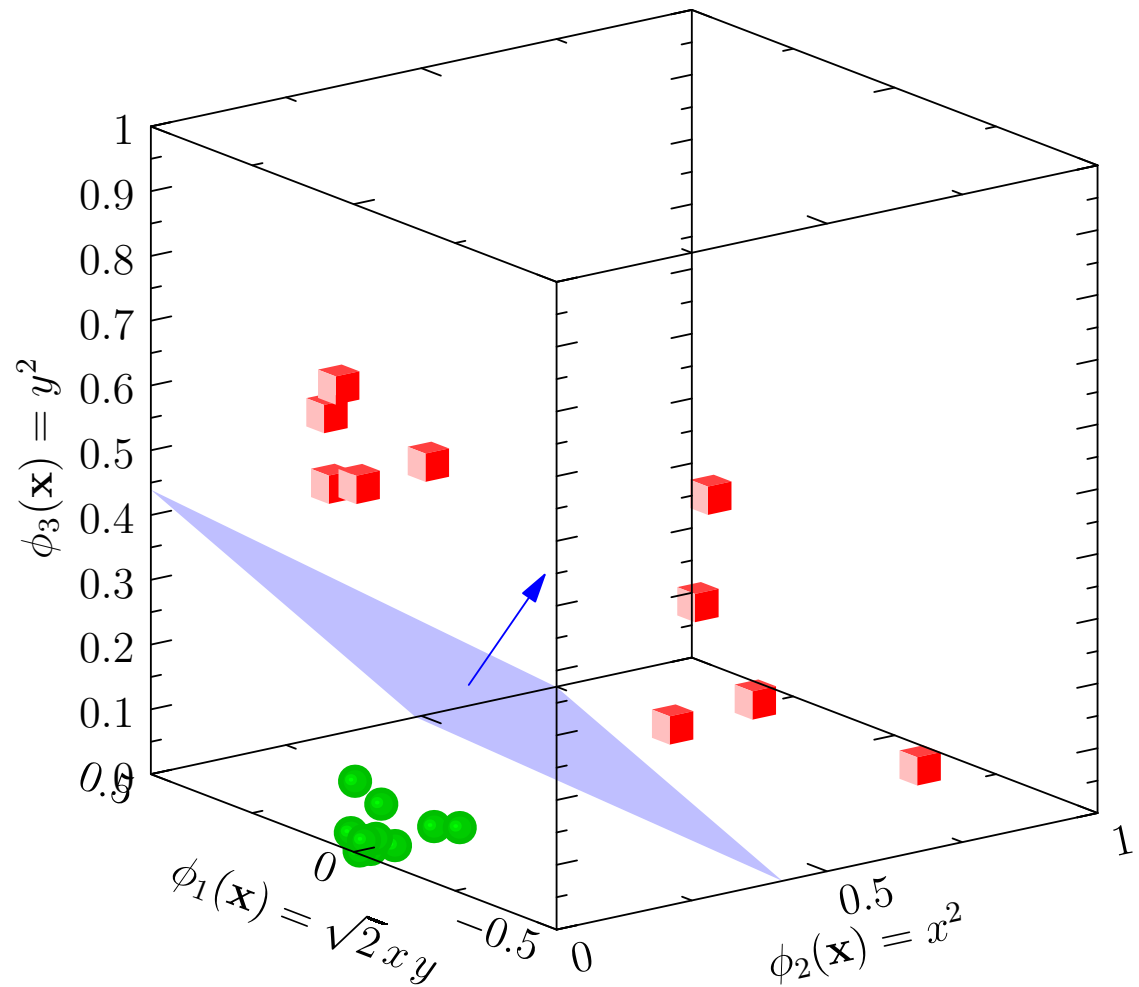
Support Vector Machines



Support Vector Machines, maximum margins

Outline

1. **The Big Picture**
2. Maximum Margins
3. Duality
4. Practice

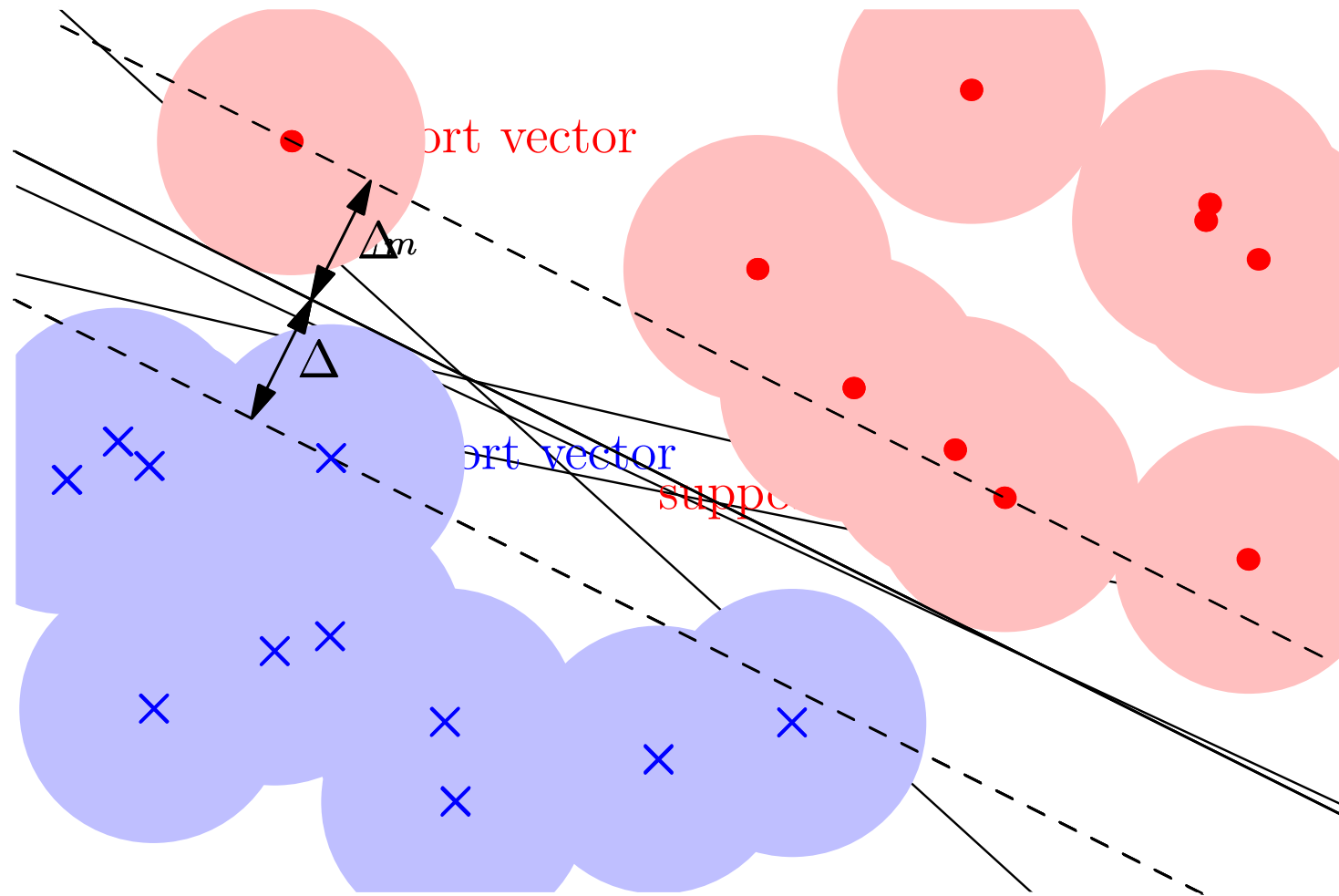


Support Vector Machines

- Support vector machines, when used right, often have the best generalisation results■
- They are typically used on numerical data, but can and have been adapted to text, sequences, etc.■
- Although not as trendy as deep learning, they will often be the method of choice on small data sets■
- They subtly regularise themselves, choosing a solution that generalises well from a host of different solutions■

Linear Separation of Data

- SVMs classify linearly separable data



- Finds maximum-margin separating plane

Extended Feature Space

- To increase the likelihood of linear-separability we often use a high-dimensional mapping

$$\mathbf{x} = (x_1, x_2, \dots, x_p)^\top \rightarrow \phi(\mathbf{x}) = (\phi_1(\mathbf{x}), \phi_2(\mathbf{x}), \dots, \phi_r(\mathbf{x}))^\top$$

$$r \gg p$$

- Finding the maximum margin hyper-plane is time consuming in “primal” form if r is large
- We can work in the “dual” space of patterns, then we only need to compute inner-products

$$\langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle = \phi(\mathbf{x}_i)^\top \phi(\mathbf{x}_j) = \sum_{k=1}^r \phi_k(\mathbf{x}_i) \phi_k(\mathbf{x}_j)$$

Kernel Trick

- If we choose a **positive semi-definite** kernel function $K(\mathbf{x}, \mathbf{y})$ then there exists functions $\phi(\mathbf{x}) = (\phi_k(\mathbf{x}) | k = 1, 2, \dots, r)$, such that

$$K(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$$

(like an eigenvector decomposition of a matrix)■

- Never need to compute $\phi_k(\mathbf{x}_i)$ explicitly as we only need the inner-product $\langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle = K(\mathbf{x}_i, \mathbf{x}_j)$ to compute maximum margin separating hyper-plane■
- Sometimes $\phi(\mathbf{x}_i)$ is an infinite dimensional vector so it is good we don't have to compute all the elements!■

Kernel Functions

- Kernel functions are symmetric functions of two variable■
- Strong restriction: *positive semi-definite*■
- Examples

Quadratic kernel: $K(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1^\top \mathbf{x}_2)^2$

Gaussian (RBF) kernel: $K(\mathbf{x}_1, \mathbf{x}_2) = e^{-\gamma \|\mathbf{x}_1 - \mathbf{x}_2\|^2}$ ■

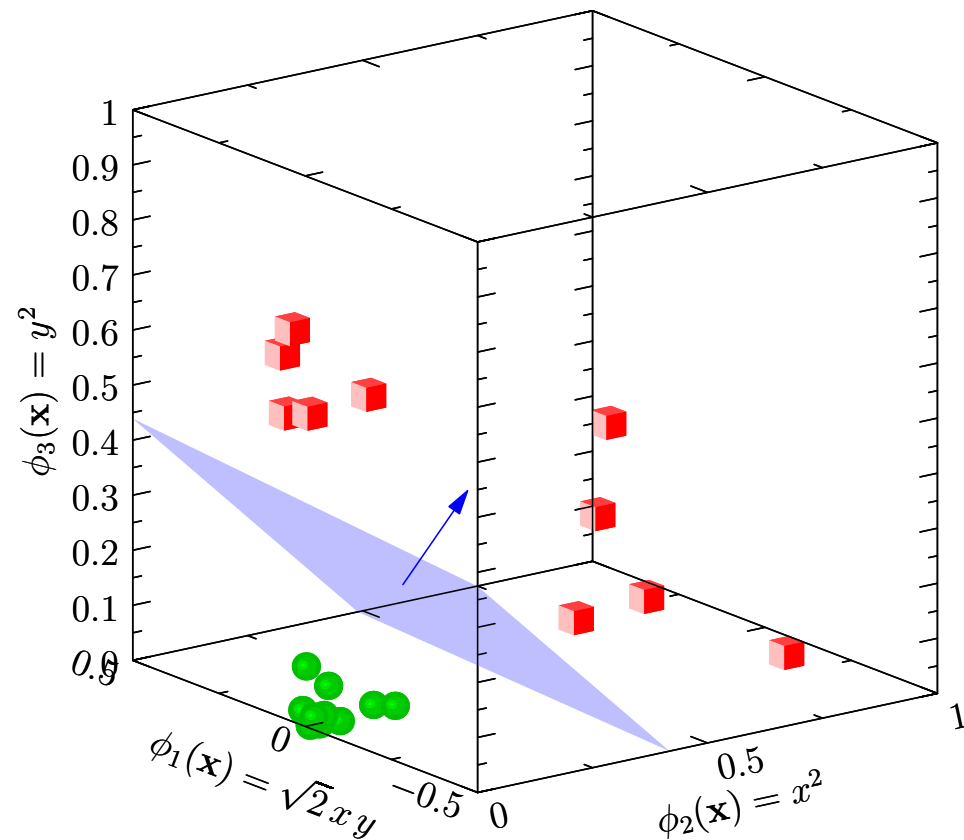
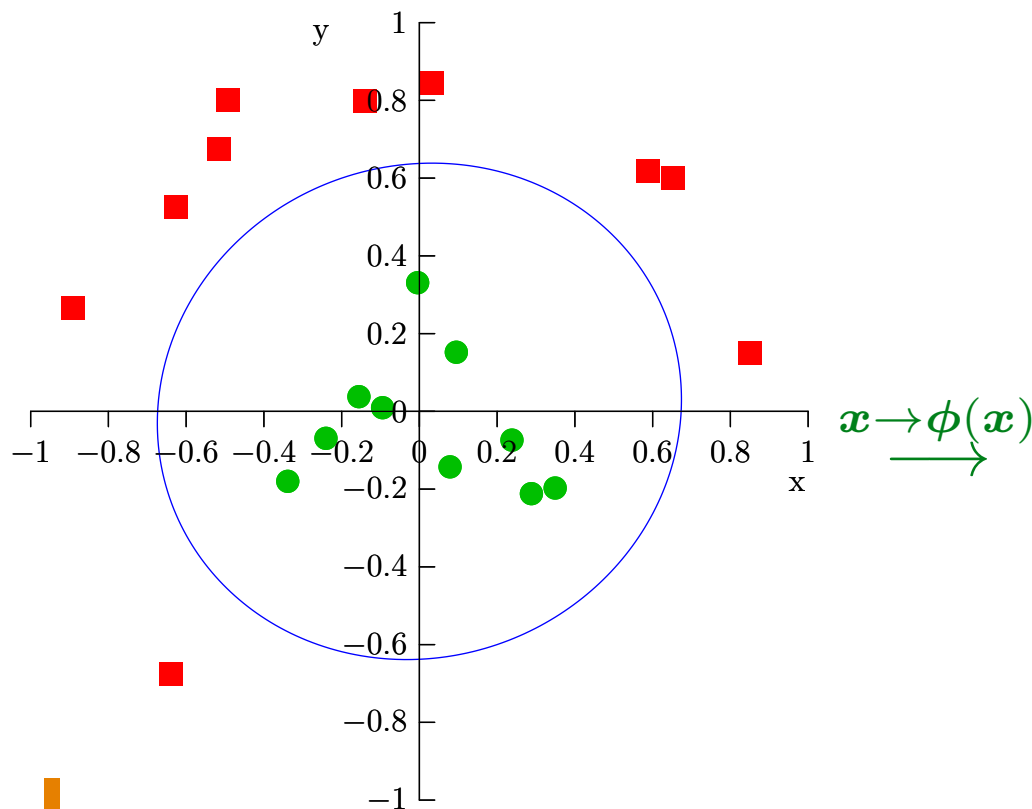
- Consider the mapping

$$\mathbf{x}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix} \rightarrow \phi(\mathbf{x}_i) = \begin{pmatrix} x_i^2 \\ y_i^2 \\ \sqrt{2}x_i y_i \end{pmatrix} \quad \blacksquare$$

Non-linear Separation of Data

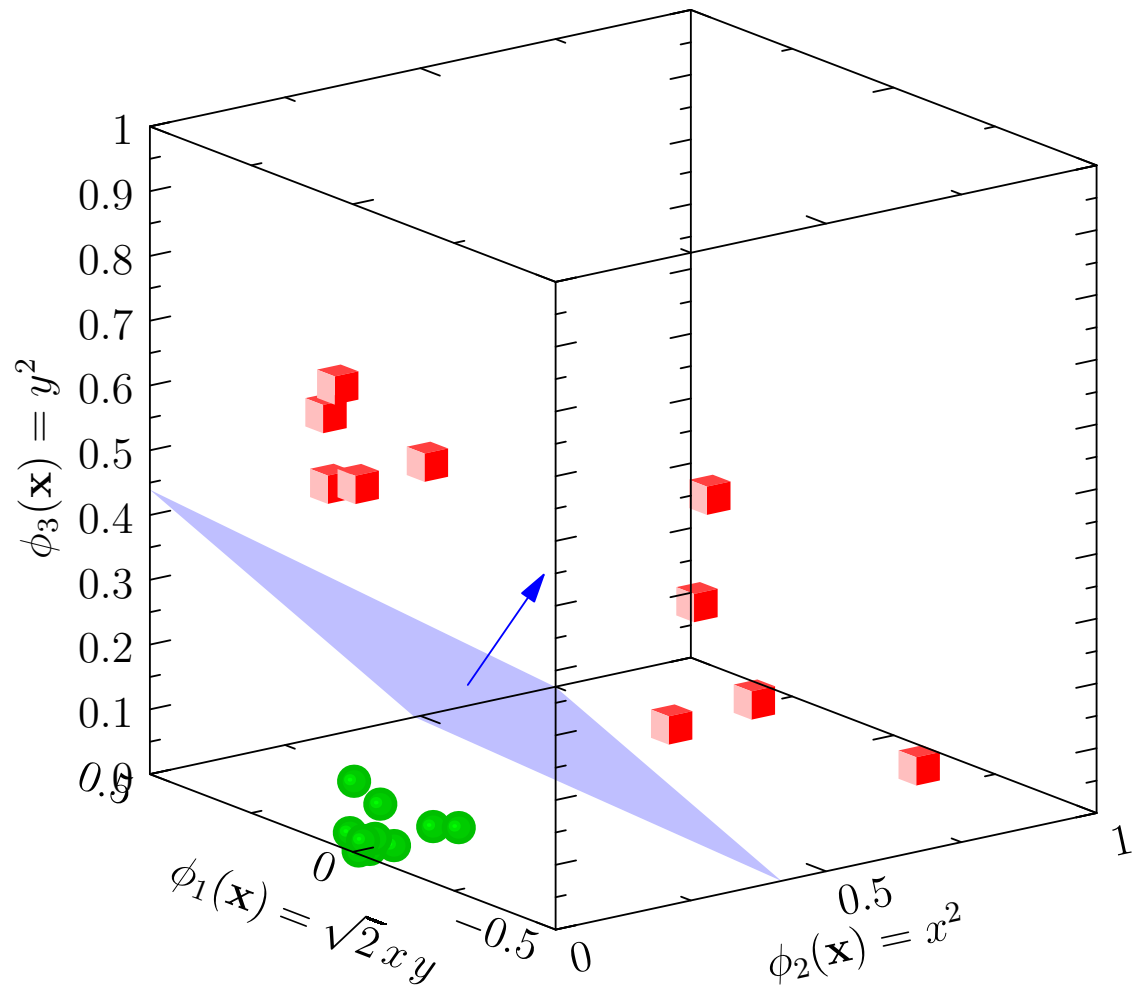
$$K(\mathbf{x}_1, \mathbf{x}_2) = \begin{pmatrix} x_1^2 & y_1^2 & \sqrt{2}x_1y_1 \end{pmatrix} \begin{pmatrix} x_2^2 \\ y_2^2 \\ \sqrt{2}x_2y_2 \end{pmatrix} = x_1^2x_2^2 + y_1^2y_2^2 + 2x_1y_1x_2y_2$$

$$= (x_1x_2 + y_1y_2)^2 = (\mathbf{x}_1^T \mathbf{x}_2)^2$$



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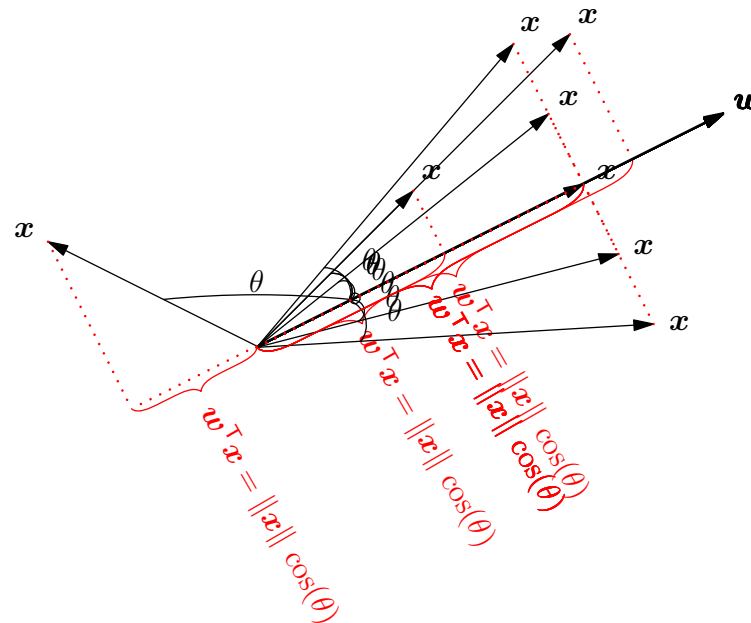


Inner Product

- Recall the inner or dot product in $\mathbb{R}^n$

$$\left(\text{---} \right) \left(\begin{pmatrix} | \\ | \\ | \end{pmatrix} \right) = \left(\begin{pmatrix} \blacksquare \end{pmatrix} \right) \quad \langle x, y \rangle = x \cdot y = x^T y = \sum_{i=1}^n x_i y_i = \|x\| \|y\| \cos(\theta)$$

- If $\|w\| = 1$ then $\langle x, w \rangle = \|x\| \cos(\theta)$



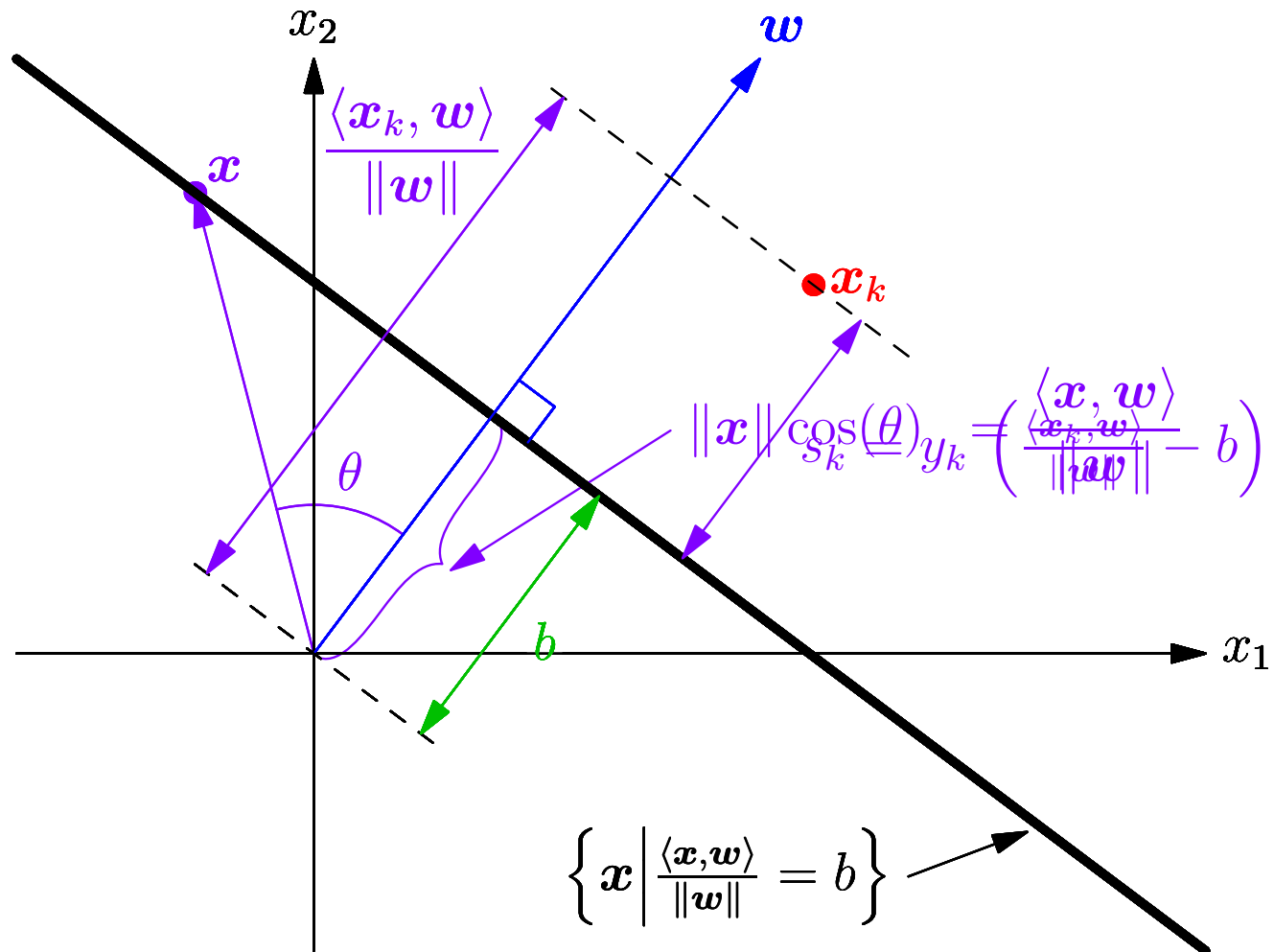
Maximise Margin

- Consider a linearly separable set of data
 - ★ $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^m$
 - ★ $y_k \in \{-1, 1\}$ ■
- Our task is to find a separating plane defined by the orthogonal vector $\mathbf{w}$ and a threshold b such that

$$y_k \left(\frac{\langle \mathbf{w}, \mathbf{x}_k \rangle}{\|\mathbf{w}\|} - b \right) \geq \Delta$$

where Δ is the margin ■

Distance to hyperplanes



Constrained Optimisation

- Wish to find $\mathbf{w}$ and b to maximise Δ subject to constraints

$$y_k \left(\frac{\langle \mathbf{w}, \mathbf{x}_k \rangle}{\|\mathbf{w}\|} - b \right) \geq \Delta \quad \text{for all } k = 1, 2, \dots, m$$

- If we divide through by Δ

$$y_k \left(\frac{\langle \mathbf{w}, \mathbf{x}_k \rangle}{\Delta \|\mathbf{w}\|} - \frac{b}{\Delta} \right) \geq 1 \quad \text{for all } k = 1, 2, \dots, m$$

- Define $\hat{\mathbf{w}} = \mathbf{w}/(\Delta \|\mathbf{w}\|)$ and $\hat{b} = b/\Delta$

$$y_k \left(\langle \hat{\mathbf{w}}, \mathbf{x}_k \rangle - \hat{b} \right) \geq 1$$

Quadratic Programming Problem

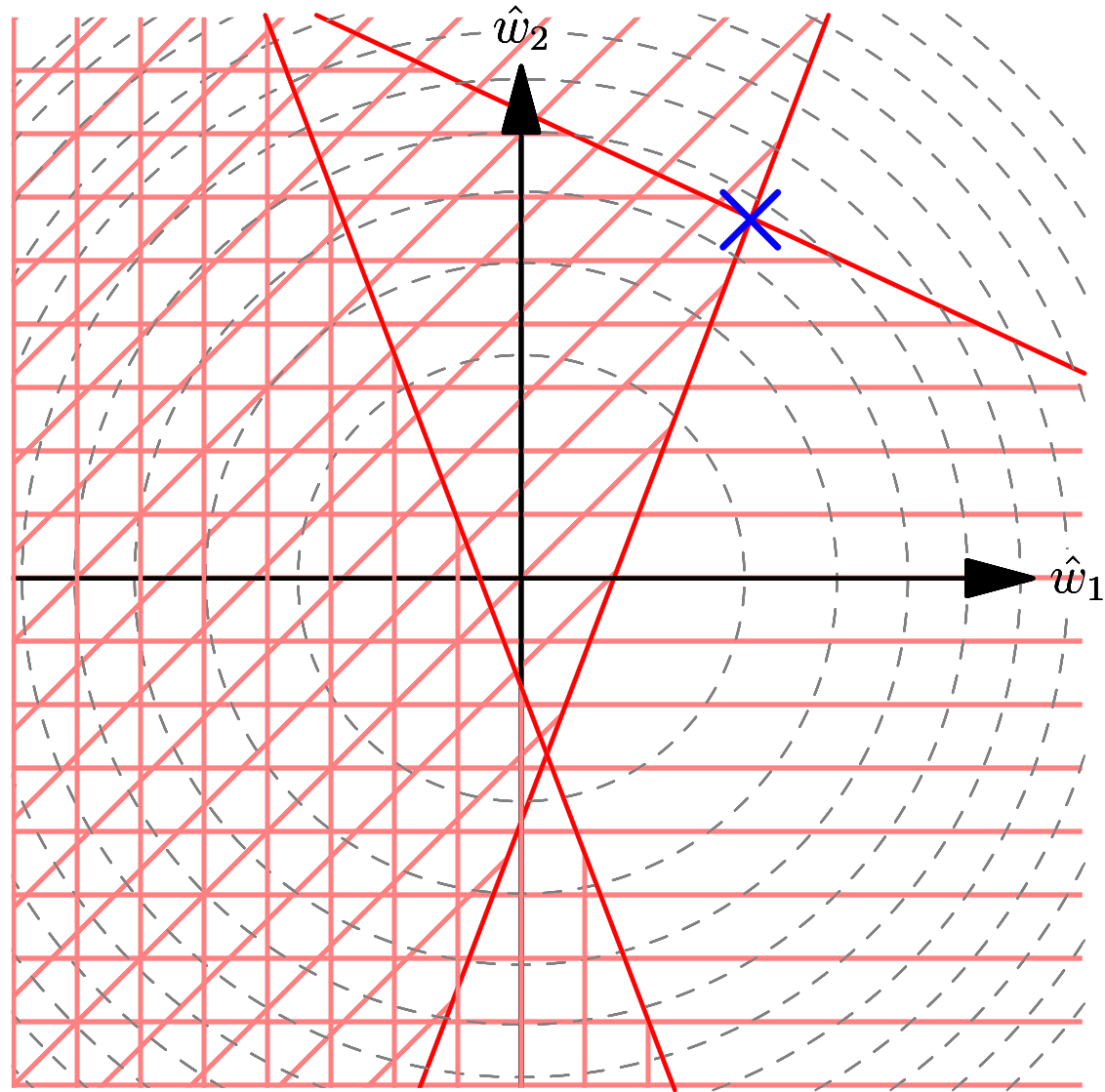
- Note that as $\hat{\mathbf{w}} = \mathbf{w} / (\Delta \|\mathbf{w}\|)$

$$\|\hat{\mathbf{w}}\| = \left\| \frac{\mathbf{w}}{\Delta \|\mathbf{w}\|} \right\| = \frac{1}{\Delta \|\mathbf{w}\|} \|\mathbf{w}\| = \frac{1}{\Delta}$$

- Minimising $\|\hat{\mathbf{w}}\|^2$ is equivalent to maximising the margin Δ
- Can write the optimisation problem as a *quadratic programming problem*

$$\min_{\hat{\mathbf{w}}, \hat{b}} \frac{\|\hat{\mathbf{w}}\|^2}{2} \quad \text{subject to } y_k \left(\langle \hat{\mathbf{w}}, \mathbf{x}_k \rangle - \hat{b} \right) \geq 1 \text{ for all } k = 1, 2, \dots, m$$

Quadratic Programming in SVMs

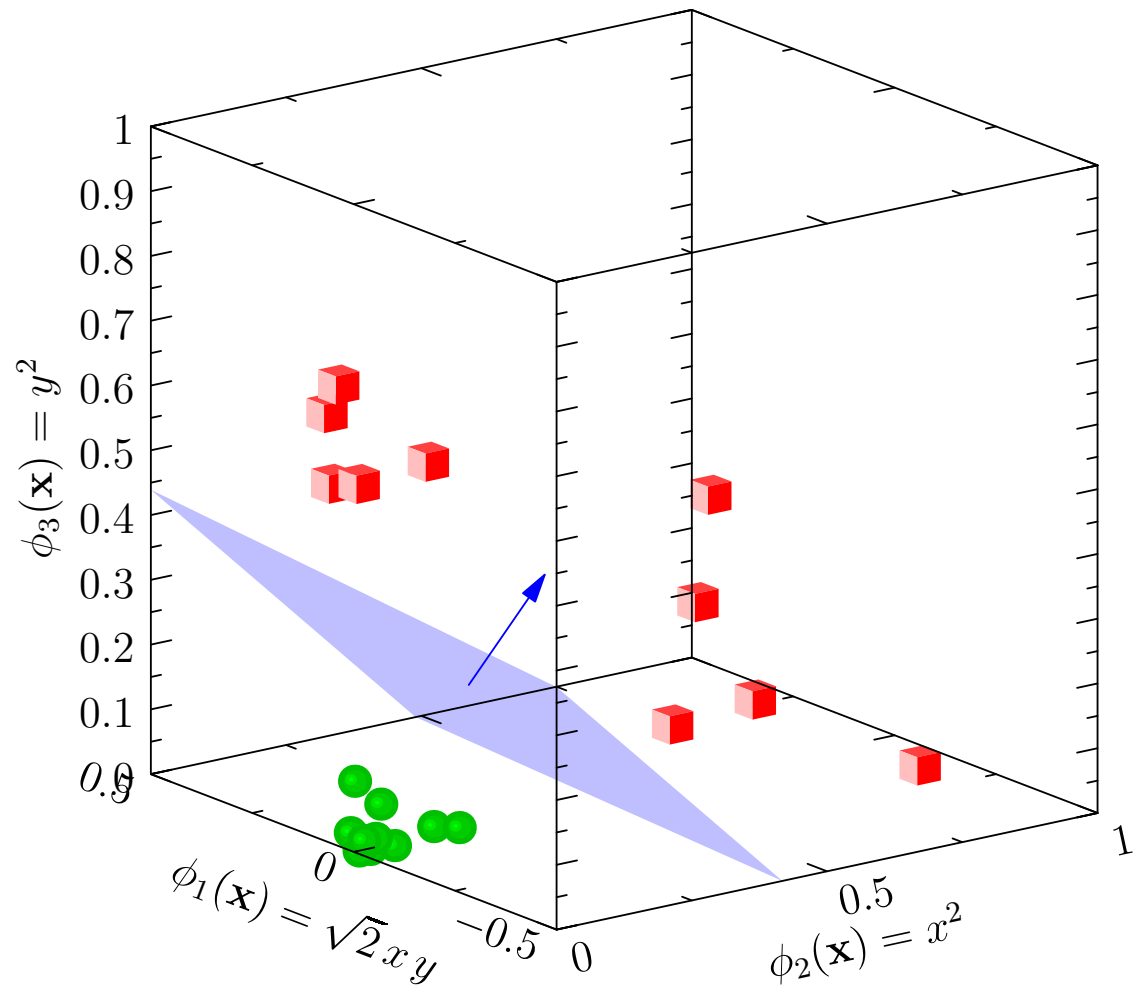


Quadratic Programming

- We have a quadratic programming problem for the weights $\hat{\mathbf{w}} = (\hat{w}_1, \hat{w}_2, \dots, \hat{w}_p)$ and bias $\hat{b}$ and m constraints■
- This is a classic but fiddly optimisation problems■
- It is a **strongly convex** optimisation problem so will have a unique minimum■
- It can be solved in $O(p^3)$ time (it involves inverting matrices)■ (phew it is not NP-complete!)■
- We will see that there is an equivalent dual problem which allows us to use the kernel trick with time complexity $O(m^3)$ ■

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Extended Feature Space

- We can generalise the SVM if we map all our features vectors to an extended feature space

$$\mathbf{x} \rightarrow \phi(\mathbf{x})$$

- The components of $\phi(\mathbf{x})$ will typically be (non-linear) functions of $\mathbf{x}$ (e.g. $\phi_1(\mathbf{x}) = x_1^2, \phi_2(\mathbf{x}) = x_2^2, \phi_3(\mathbf{x}) = \sqrt{2}x_1x_2$)
- We are free to choose whatever mappings we like
- There may be many more components of $\phi(\mathbf{x})$ than of $\mathbf{x}$ making it easier to find a linear separation of the two classes
- But in the extended feature space (involving $\phi(\mathbf{x}) = (\phi_1(\mathbf{x}), \phi_2(\mathbf{x}), \dots, \phi_r(\mathbf{x}))$) the time complexity is $O(r^3)$

Lagrangian

- In the extended feature space we can find a separating plane (given by $\mathbf{w}$ and b) with maximum margin by solving the problem

$$\min_{\mathbf{w}, b} \frac{\|\mathbf{w}\|^2}{2} \quad \text{subject to } y_k(\langle \mathbf{w}, \phi(\mathbf{x}_k) \rangle - b) \geq 1 \text{ for all } k = 1, 2, \dots, m$$

- We can write this as a Lagrange problem

$$\min_{\mathbf{w}, b} \max_{\alpha \geq 0} \mathcal{L}(\mathbf{w}, b, \alpha)$$

where

$$\mathcal{L}(\mathbf{w}, b, \alpha) = \frac{\|\mathbf{w}\|^2}{2} - \sum_{k=1}^m \alpha_k \left(y_k (\langle \mathbf{w}, \phi(\mathbf{x}_k) \rangle - b) - 1 \right)$$

subject to $\alpha_k \geq 0$

Obtaining the Dual Form of the Problem

- Differentiating the Lagrangian with respect to $\mathbf{w}$

$$\mathcal{L}(\mathbf{w}, b, \alpha) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{k=1}^m \alpha_k \left(y_k (\langle \mathbf{w}, \phi(\mathbf{x}_k) \rangle - b) - 1 \right)$$

- $\nabla_{\mathbf{w}} \mathcal{L} = \mathbf{w} - \sum_{k=1}^m \alpha_k y_k \phi(\mathbf{x}_k) = 0 \implies \mathbf{w}^* = \sum_{k=1}^m \alpha_k y_k \phi(\mathbf{x}_k)$

- $\frac{\partial \mathcal{L}}{\partial b} = \sum_{k=1}^m \alpha_k y_k = 0$

- Substituting back into the Lagrangian

$$\max_{\alpha \geq 0} \mathcal{L}(\mathbf{w}^*, b^*, \alpha) = \max_{\alpha \geq 0} \sum_{k=1}^m \alpha_k - \frac{1}{2} \sum_{k,l=1}^m \alpha_k \alpha_l y_k y_l \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$$

The Dual Problem

- The dual problem is now to find α_k 's that maximise

$$\mathcal{L}(\alpha) = \sum_{k=1}^m \alpha_k - \frac{1}{2} \sum_{k,l=1}^m \alpha_k \alpha_l y_k y_l \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle \blacksquare$$

- subject to constraints

$$\sum_{k=1}^m \alpha_k y_k = 0 \quad \forall k = 1, 2, \dots, m \quad \alpha_k \geq 0 \blacksquare$$

- The Hessian of $\mathcal{L}(\alpha)$ has elements $H_{kl} = -y_k y_l \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$ so $\mathbf{v}^\top \mathbf{H} \mathbf{v} = -\|\sum_k v_k y_k \phi_k(\mathbf{x}_k)\|^2 \leq 0$ (note this is negative semi-definite so there is a unique maximum) $\blacksquare$

Kernel Trick

- We will show in the next lecture that if $K(\mathbf{x}, \mathbf{y})$ is a positive semi-definite function then it can always be written as

$$K(\mathbf{x}, \mathbf{y}) = \langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle \blacksquare$$

- As $\langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$ appears in the dual problem we can express the dual problem as finding α_k 's that maximise

$$\mathcal{L}(\boldsymbol{\alpha}) = \sum_{k=1}^m \alpha_k - \frac{1}{2} \sum_{k,l=1}^m \alpha_k \alpha_l y_k y_l K(\mathbf{x}_k, \mathbf{x}_l) \blacksquare$$

- We therefore never have to compute $\phi(\mathbf{x})$ $\blacksquare$

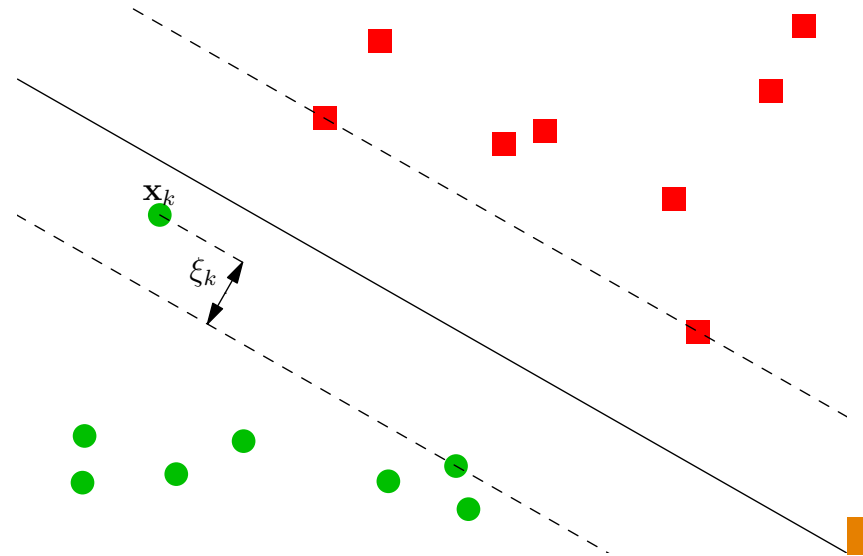
Sequential Minimal Optimisation

- One of the most efficient techniques for training SVMs is *Sequential Minimal Optimisation* or SMO■
- This takes two Lagrange multipliers α_i and α_j and adjusts them to maximise the dual objective function■
- This is very quick as it can be done in closed form■
- Note that because $\sum_{k=1}^m y_k \alpha_k = 0$ we have to change at least two variables at the same time■
- A heuristic is used to choose good pairs of α 's to optimise■
- Run until close to the optimum■

Soft Margins

- We can relax the margin constraints by introducing *slack variables*, $s_k \geq 0$

$$y_k(\langle \mathbf{x}_k, \mathbf{w} \rangle - b) \geq 1 - s_k$$



- Minimise $\frac{\|\mathbf{w}\|^2}{2} + C \sum_{k=1}^n s_k$
- Larger C punishes slack variables more

Dual Problem with Slack Variables

- The Lagrangian with slack variables is

$$\mathcal{L} = \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{k=1}^m s_k - \sum_{k=1}^m \alpha_k \left(y_k (\langle \mathbf{w}, \phi(\mathbf{x}_k) \rangle - b) - 1 + s_k \right) - \sum_{k=1}^m \beta_k s_k$$

where β_k are Lagrange multipliers that ensure $s_k \geq 0$ (note that $\beta_k \geq 0$ —this is the KKT condition)■

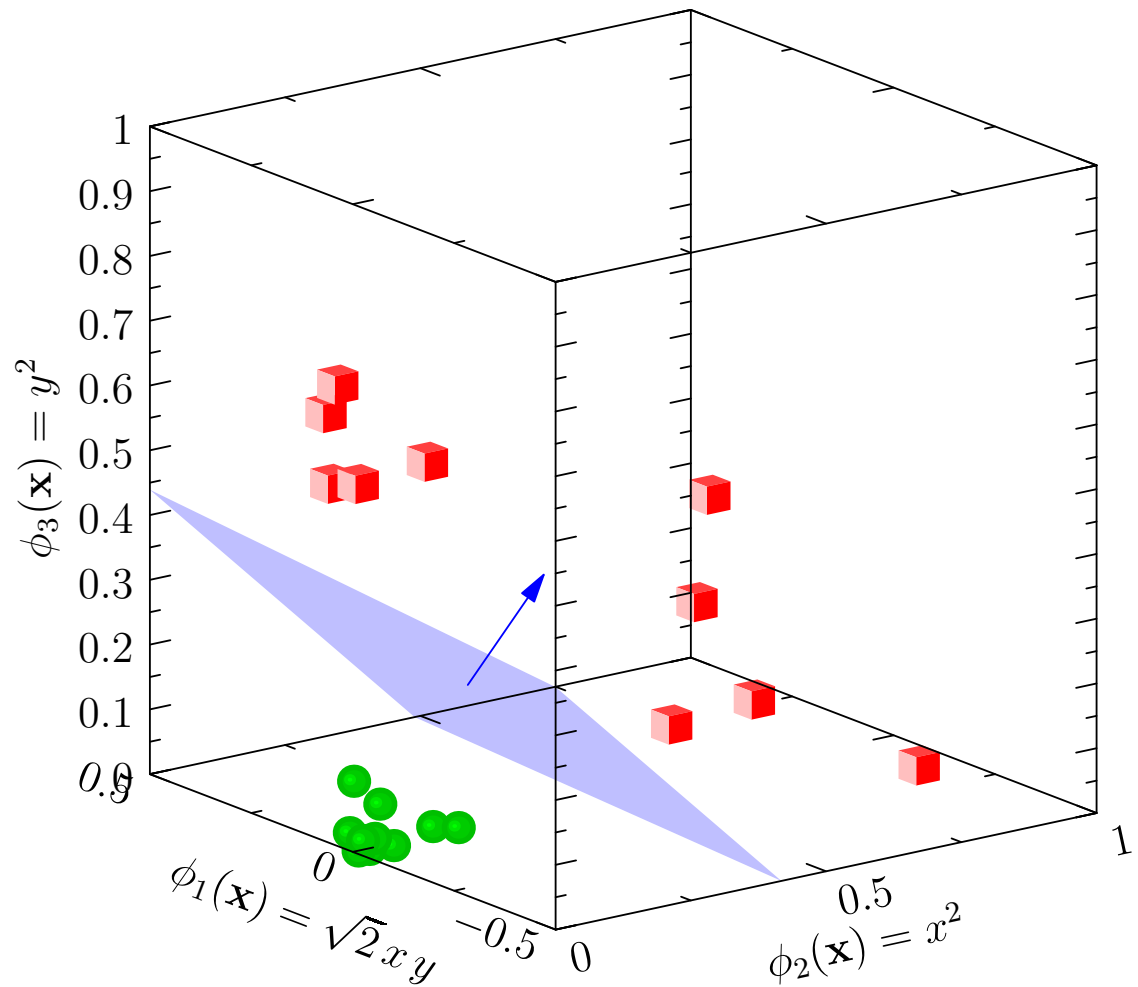
- Now minimising with respect to s_i

$$\frac{\partial \mathcal{L}}{\partial s_i} = C - \alpha_i - \beta_i = 0 \blacksquare$$

- Or $\alpha_i = C - \beta_i$. Since $\beta_i \geq 0$ the constraint is $\alpha_i \leq C$ ■ (recall also $\alpha_i \geq 0$)■

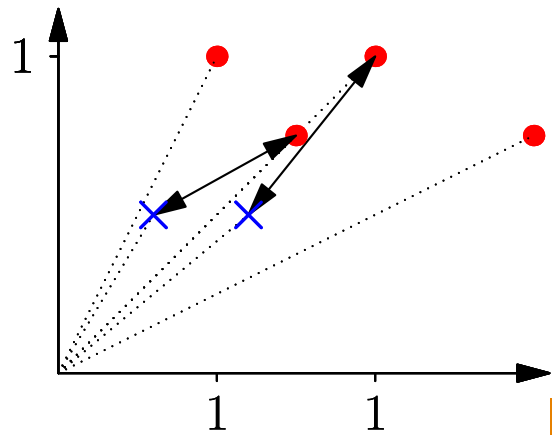
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Getting SVMs to Work Well

- SVMs rely on distances between data points
- These will change relative to each other if we rescale some features but not other—giving different maximum-margin hyper-planes



- If we don't know what features are important (most often the case), then it is worth scaling each feature (for example, so their range is between 0 and 1 or their variance is 1)

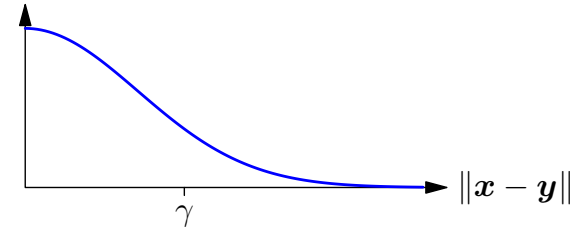
Optimising C

- Recall that we can introduce soft-margins using slack variables where we minimise $\frac{\|\hat{\mathbf{w}}\|^2}{2} + C \sum_{k=1}^m s_k$ subject to constraints
- In practice it can make a huge difference to the performance if we change C
- Optimal C values changes by many orders of magnitude e.g. 2^{-5} – 2^{15}
- Typically optimised by a grid search (start from 2^{-5} say and double until you reach 2^{15})
- Measure performance on a validation set

Choosing the Right Kernel Function

- There are kernels design for particular data types (e.g. string kernels for text or biological sequences)■
- For numerical data, people tend to look at using no kernel (linear SVM), a radial basis function (Gaussian) kernel or polynomial kernels■
- Kernels often come with parameters, e.g. the popular radial basis function kernel

$$K(\mathbf{x}, \mathbf{y}) = e^{-\gamma \|\mathbf{x} - \mathbf{y}\|^2} \blacksquare$$



- Optimal γ values range over 2^{-15} – 2^3 ■

Multi-Class Problems

- By construction SVMs separate only two classes■
- If we have a multi-class problem we have to use multiple SVMs■
- There are two major ways practitioners do this
 - One-versus-all:** for each class, train a separate classifier to determine that class versus all others■
 - All-pairs:** train a classifier for all pairs of classes■
- In both cases choose the class which the classifier is most certain about■
- Beware SVMs don't like imbalanced datasets■

SVM Libraries

- Although SVMs have unique solutions, they require very well written optimisers■
- If you have a large data set they can be very slow■
- There are good libraries out there: svmlib, svm-lite, (now old), scikit-learn, etc.■
- These will often automate normalisation of data and grid search for parameters■

Conclusions

- We've seen how SVMs work■
- We've learnt how to use them■
- We've seen that we can find the maximum margin hyper-plane by solving a quadratic programming problem (with a unique solution)■
- This is a convex optimisation problem with a unique optimum■
- The **dual problem** of an SVM is particularly simple, especially if we use a positive semi-definite kernel■ (we explore these in the next lecture)■